

Part I. Calculus I competency.

1. (5 pts) Find the equation of the tangent line to the graph of the function $f(x) = \sqrt{2x^2 - 7}$ at $x = 4$.
2. (5 pts) Find the derivative of $a(x) = x^{5/3} - \frac{4}{x^7}$
3. (5 pts) Find the derivative of $b(x) = \frac{3x+2}{4-5x}$
4. (5 pts) Find the derivative of $c(x) = e^{-7x} \cos 6x$
5. (5 pts) Find the derivative of $e(x) = x^2 \ln x - 2x$
6. (5 pts) Find the derivative of $f(x) = (x^2 + x - 1)^7$
7. (5 pts) Find the indefinite integral $\int (10x^7 - \sqrt[4]{x^3}) dx$
8. (5 pts) Find the definite integral of $\int_0^{\pi/2} (\cos x - \sin x) dx$
9. (5 pts) Find the indefinite integral $\int (3 - 4x)^7 dx$
10. (5 pts) Find the indefinite integral $\int \frac{1 - \sqrt{\ln x}}{x} dx$

Part II. Calculus II Competency.

11. (5 pts) Find the area the bounded in the first quadrant bounded between the graphs of $y = x^2 + 2x$ and $y = 4x$.
12. (5 pts) Find the indefinite integral $\int x^2 e^x dx$
13. (5 pts) Find the partial fraction decomposition for the function $f(x) = \frac{3x-1}{x^2-4x}$
14. (5 pts) Find the indefinite integral $\int \frac{2x-1}{\sqrt{x-x^2}} dx$
15. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} \frac{k}{2k^2+1}$
16. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} k^2 \left(\frac{4}{7}\right)^{k+1}$
17. (5 pts) Find the first four non-zero terms of the MacLaurin series of $f(x) = \ln(1-x^2)$

Part III. Calculus III competency.

18. (5 pts) Find f_x, f_y for $f(x, y) = xe^{x^2y}$

19. (5 pts) Find f_{xx}, f_{yx} for $f(x, y) = xe^{x^2y}$

20. (5 pts) Find f_x, f_y, f_z for $f(x, y, z) = \frac{x + \sqrt{y - z}}{z^3}$